

Time-series Analytics Decomposition & Detrending

Giacomo Ziffer

giacomo.ziffer@polimi.it

PhD student @ Politecnico di Milano

Founder & CEO @ motus ml



POLITECNICO
MILANO 1863

Stationarity
is a good
property for
a time series



Photo by [cottonbro](#) from [Pexels](#)

However

- Very few time series in the wild are stationary
- Normally, they include
 - a long-term trend (including aperiodic cycles) and
 - one or more seasonal patterns
- We learnt to manually subtract trend and seasonal patterns hoping for the residual to be stationary
- But, what's the theory?



Let's change
perspective



Photo by [Chokniti Khongchum](#) from [Pexels](#)

Decomposition & Detrending

Thinking ...

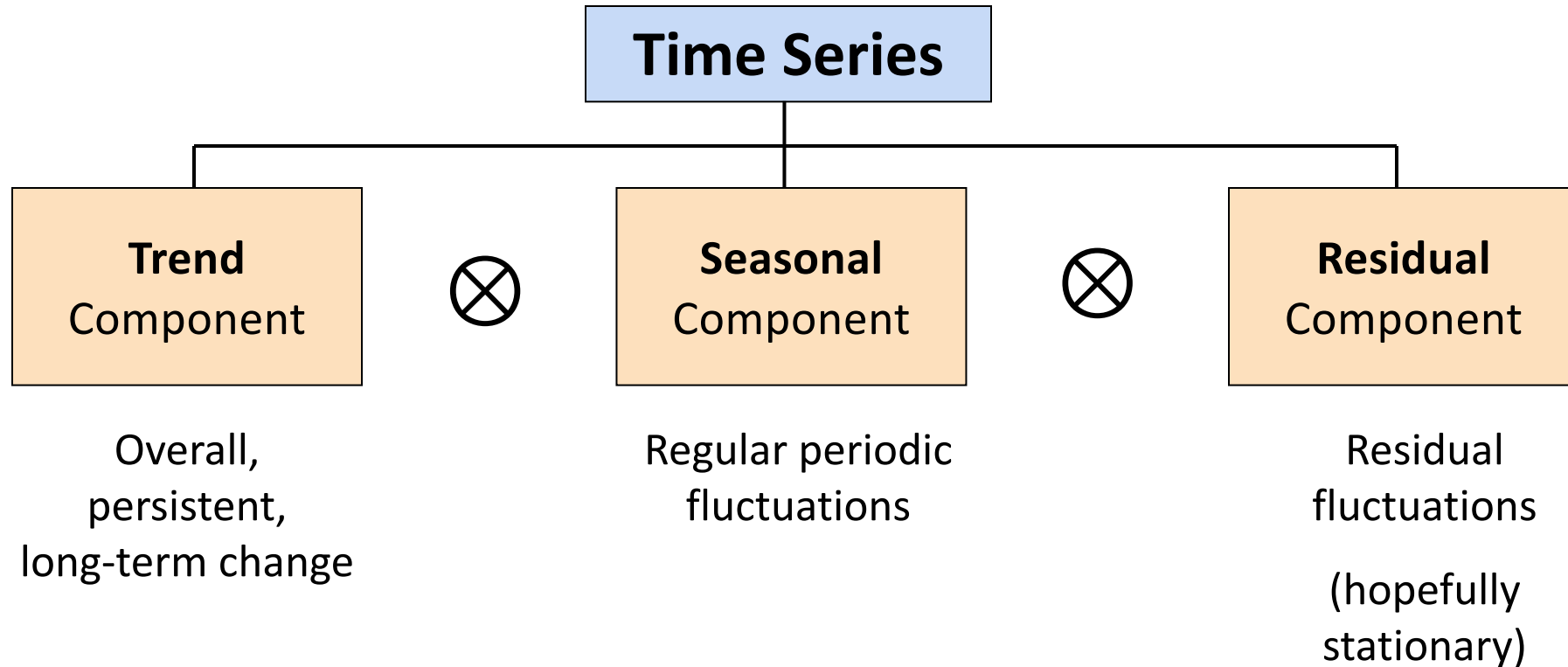
- If
 - stationary **implies** predictable
- and
 - changes in mean, variance
 - presence of seasonality
 - **imply** non-stationary
- **can we try to remove what causes of non-stationarity?**

Thinking ...

- If
 - stationary **implies** predictable
- and
 - changes in mean, variance
 - presence of seasonality
 - **imply** non-stationary
- **can we try to remove what causes of non-stationarity?**
- **YES! Decomposing a time series**

Time Series Decomposition

Time-Series Components



 this character is a placeholder for various mathematical operation used to assemble the components

Models to decompose time series:

- **Additive** model

$$X_t = m_t + s_t + Y_t$$

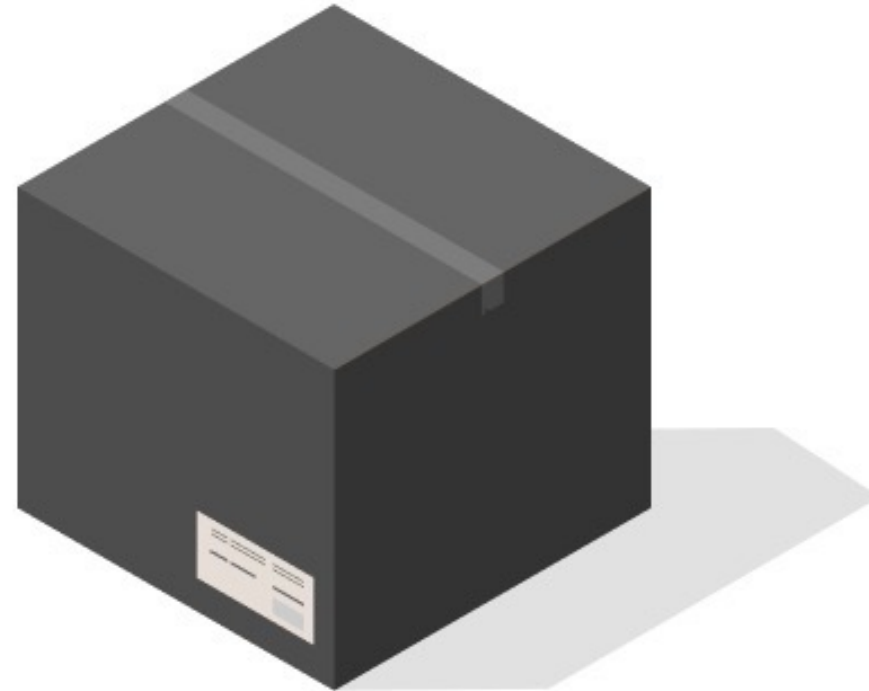
- **Multiplicative** model

$$X_t = m_t s_t Y_t$$

- Where

- m_t is the trend component
- s_t is the seasonal component
- Y_t is the residual component

Let's first go
black-box



**Black box – we do not
know anything**



- Use statsmodels as a *black-box* able to decompose a time-series
- Try both methods
 - Additive
 - Multiplicative

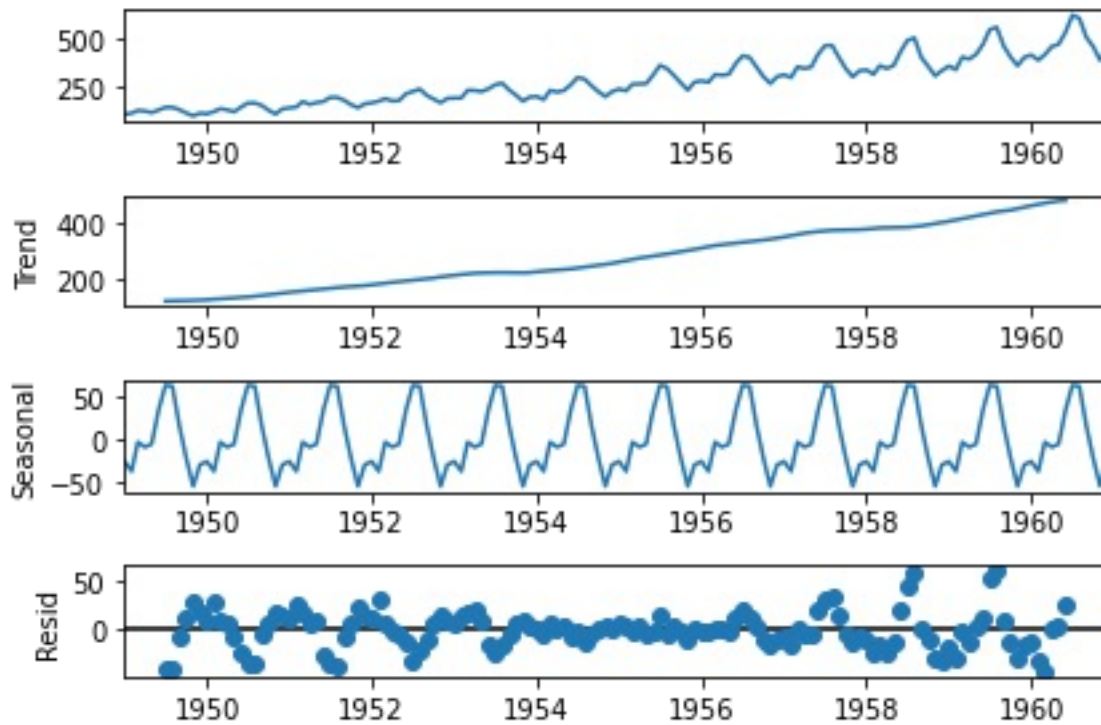


NOTE: for more information check out https://www.statsmodels.org/stable/examples/notebooks/generated/statespace_seasonal.html

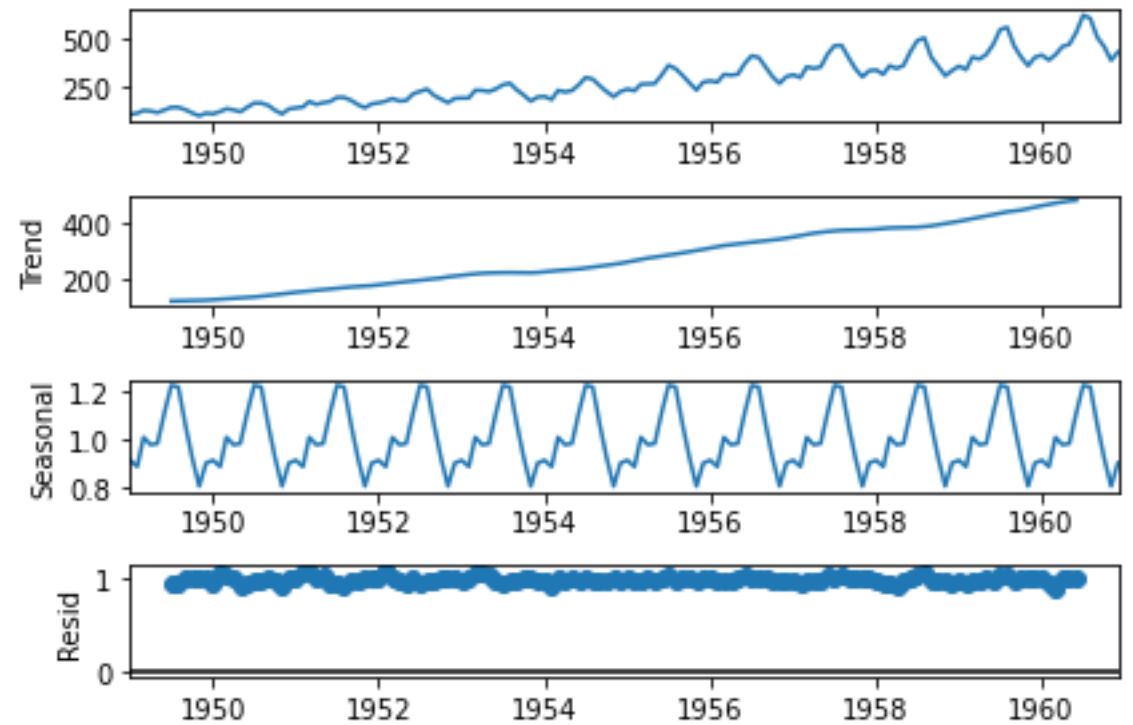
Time series decomposition

A comparison

Additive



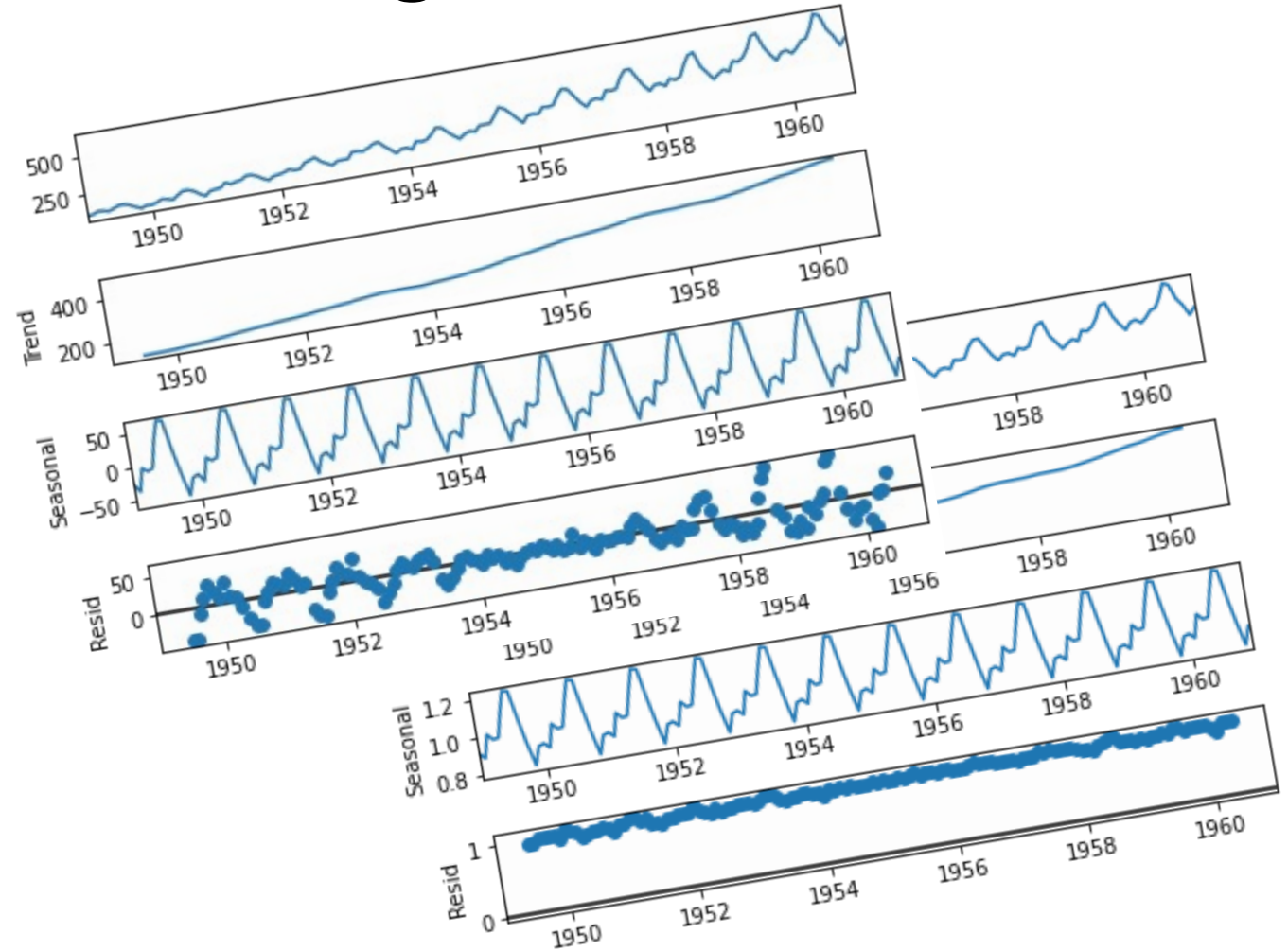
vs. Multiplicative



Time series decomposition

Let's see if you are following ...

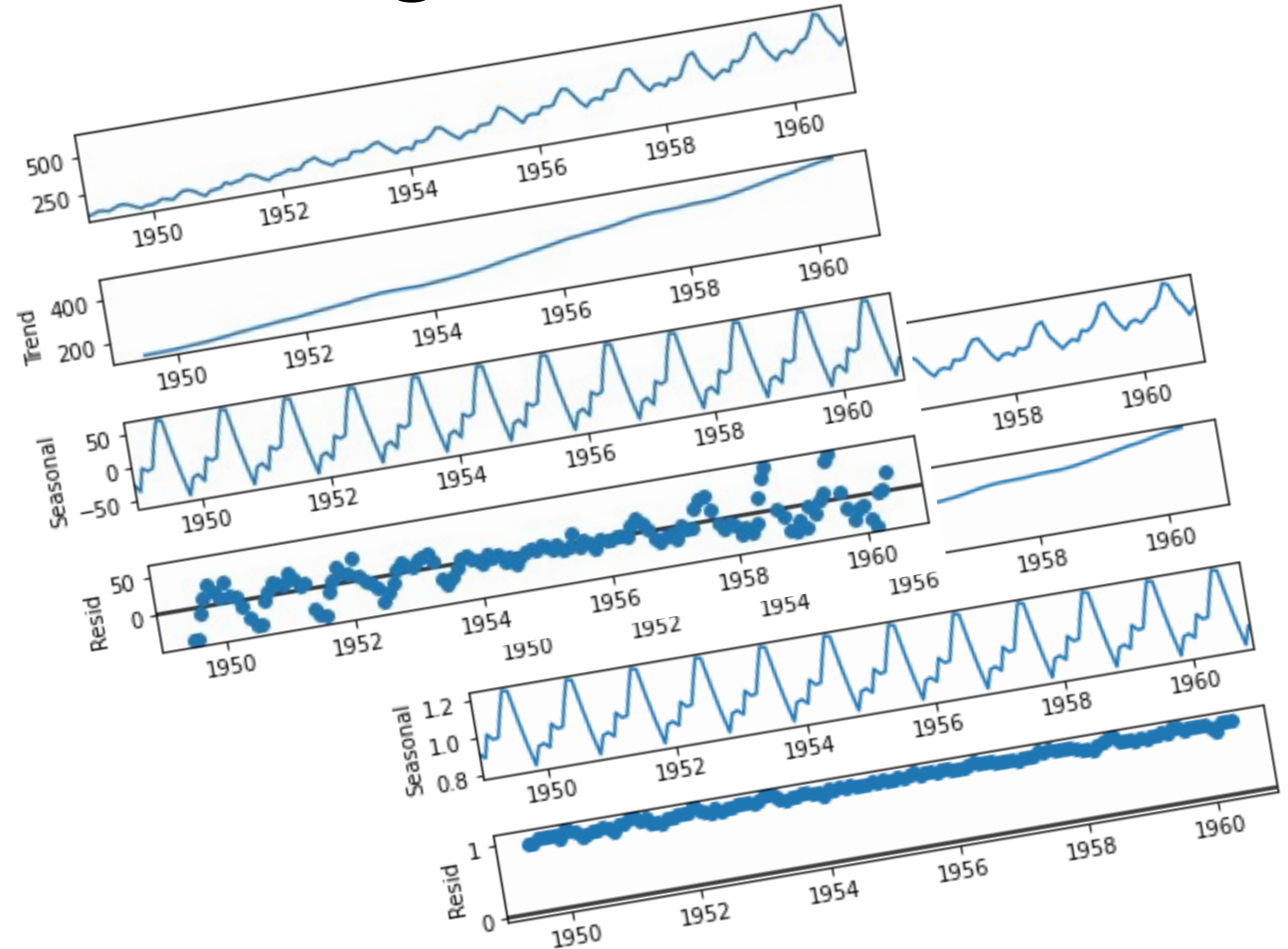
- Q: Is additive better than multiplicative?
- Q: Why?



Time series decomposition

Let's see if you are following ...

- Q: Is additive better than multiplicative?
- A: no, the multiplicative is better
- Q: Why?
- The residual of the additive decomposition does not have constant variance



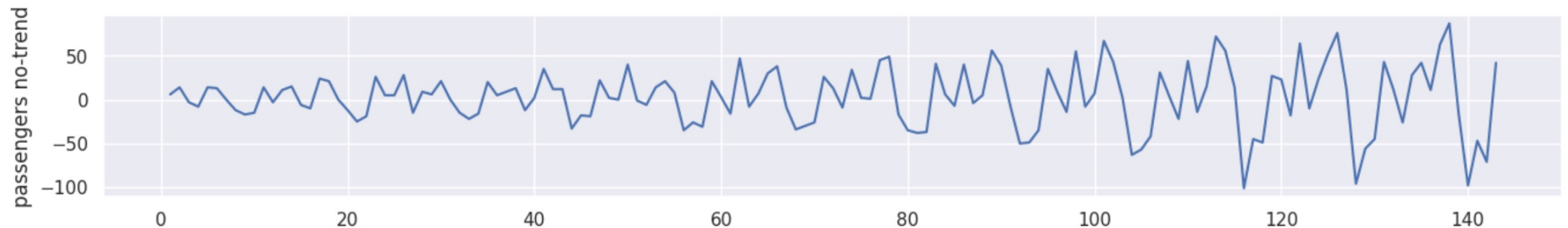
How to Choose Between ...

Additive Decompositions

- The additive model is useful when the **seasonal pattern** is relatively **independent of the trend**
- **E.g.**, when the seasonal pattern is **constant over time**

Multiplicative Decompositions

- The multiplicative model is useful when the **seasonal pattern** is **correlated with the trend**
- **E.g.**, **increases over time** when the trend increases



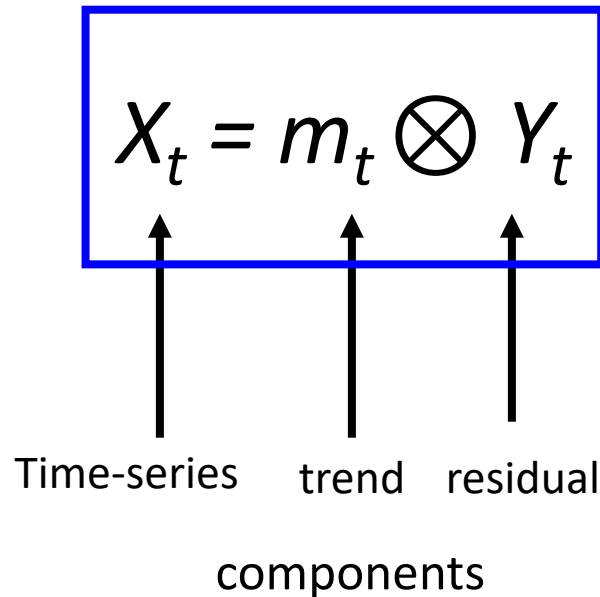
Let's now go
white-box



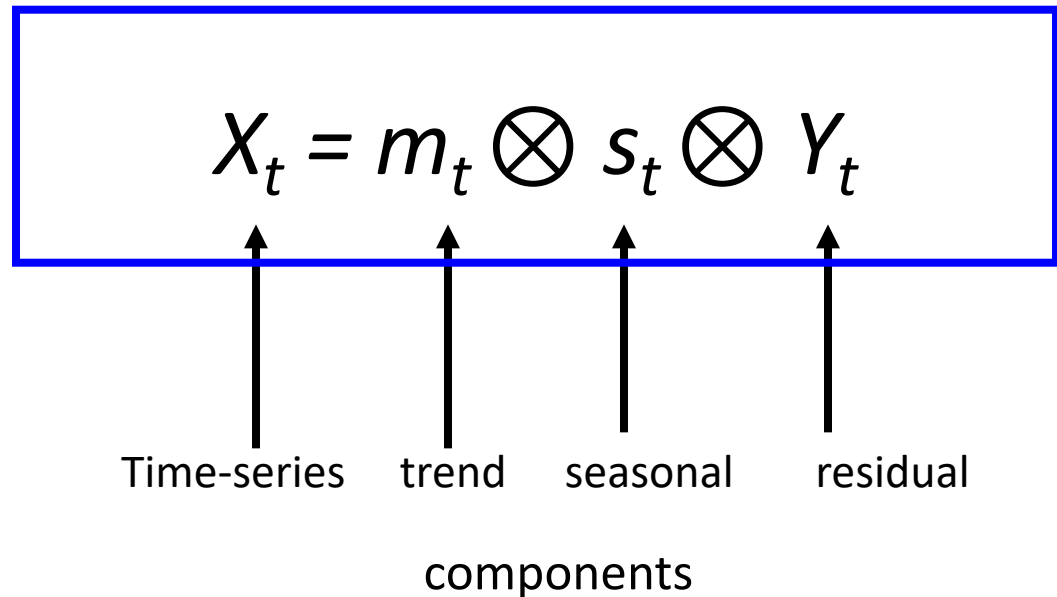
**White box - we know
everything**

Two simplified time series models

Non-seasonal Decomposition Model with Trend



Decomposition Model **with** Trend and **Seasonal** Components



Non-seasonal Decomposition Models with Trend

There are three basic methods for estimating/removing trend:

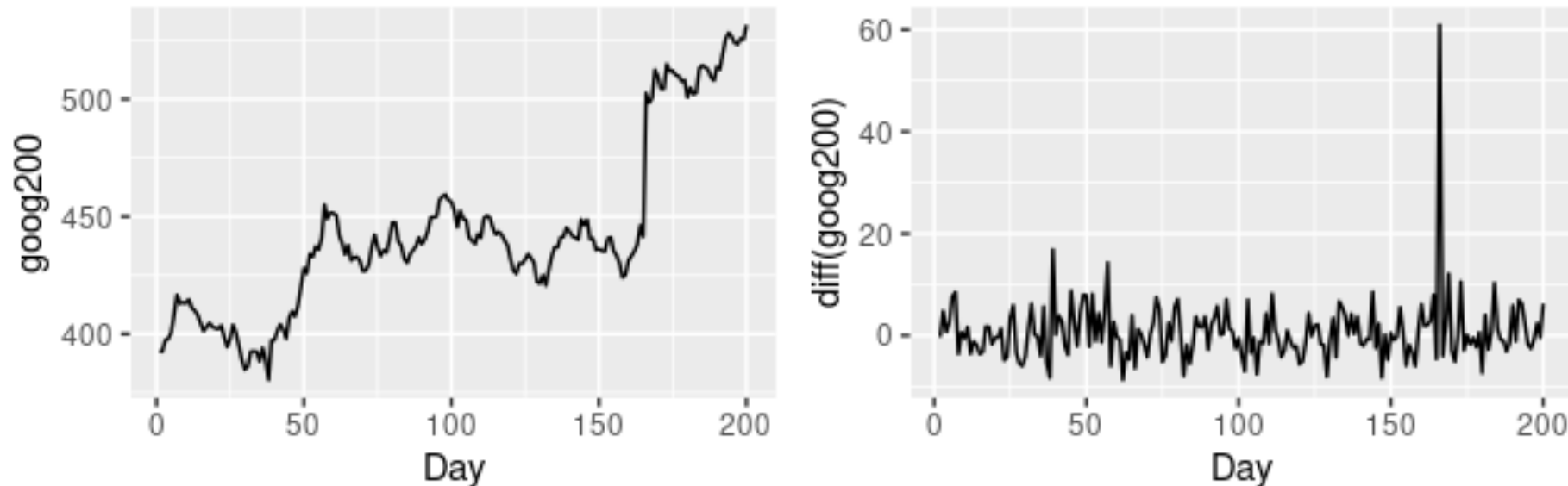
- Method 1: *Trend elimination by differencing*
 - Differencing one or more times removes the trend
- Method 2: *Trend estimation by model fitting & removal*
 - We estimate the trend fitting a model and then we remove it
- Method 3: *Trend estimation by using “centered” moving averages*
 - We estimate the trend using a “centered” moving average and we remove it

Non-seasonal Decomposition Models with Trend - Method 1

Trend elimination by differencing

- Differencing of a time series $\{X_t\}$ in discrete time t is the transformation of the series to a new time series $\{D_t\}$ where the values are the differences between consecutive values of $\{X_t\}$.

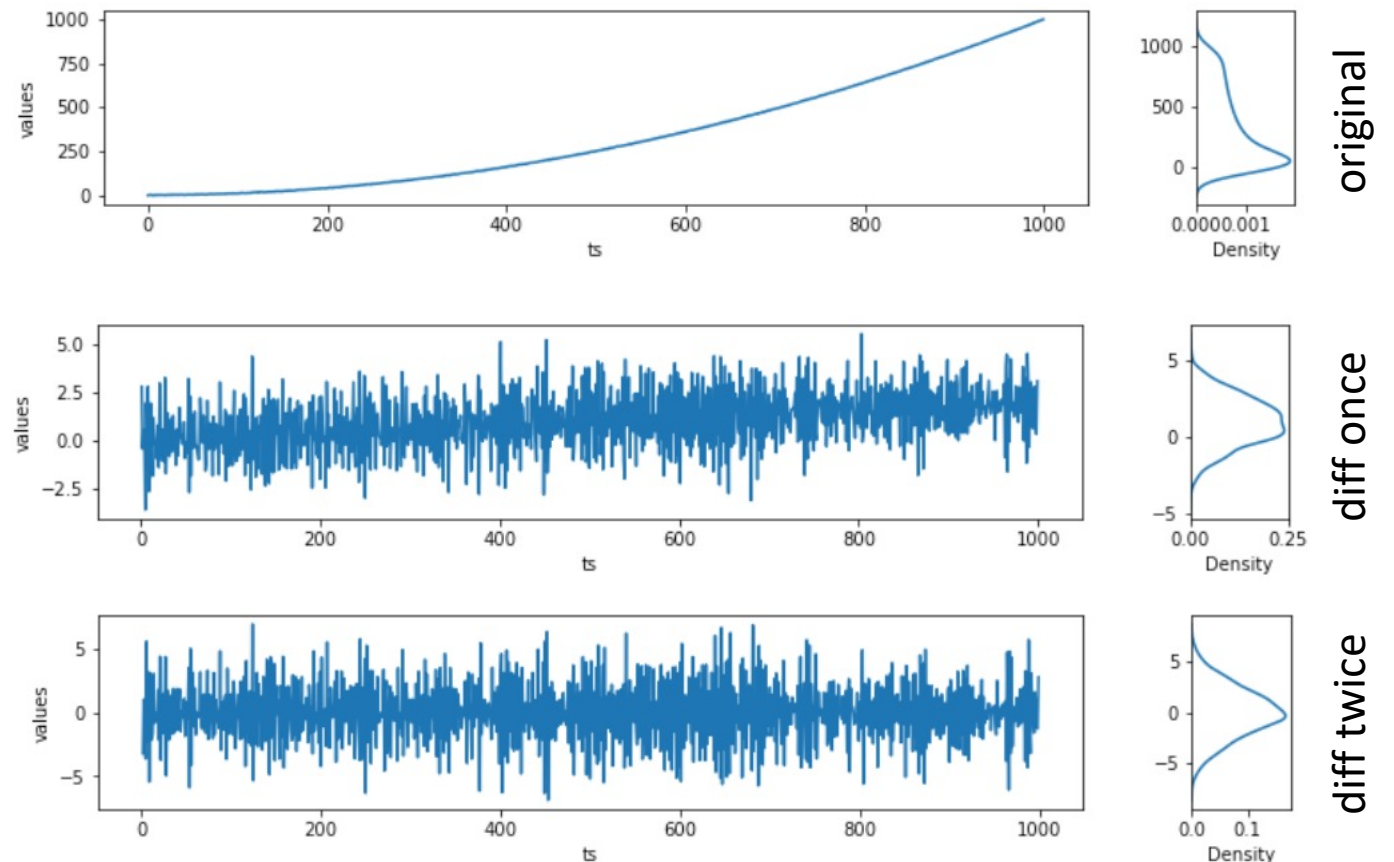
$$d_t = x_t - x_{t-1}$$



Non-seasonal Decomposition Models with Trend - Method 1

Trend elimination by differencing (cont.)

- If a trend is linear differencing once is sufficient to remove it
- If a trend is quadratic, you need two difference twice
- If a trend can be model with a polynomial of order n , then you need to difference n times



Non-seasonal Decomposition Models with Trend - Method 1

Trend elimination by differencing

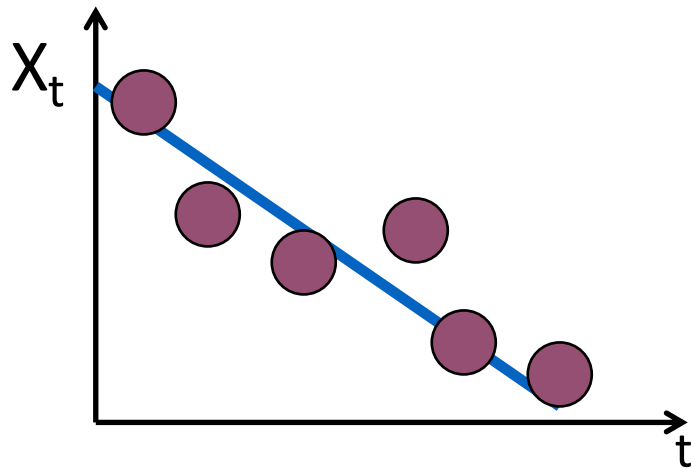
1. Generate synthetic data
2. Differencing one to remove a linear trend
3. Differencing twice to remove a quadratic trend



Non-seasonal Decomposition Models with Trend - Method 2

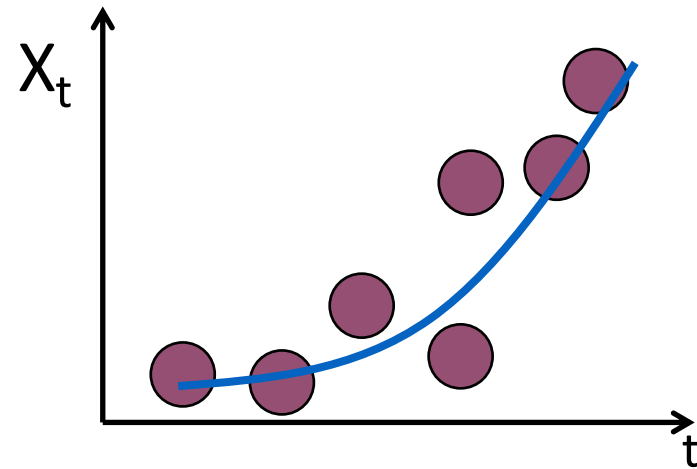
Trend estimation by model fitting & removal

$$X_t = b_0 + b_1 t$$



Downward linear trend

$$X_t = b_0 + b_1 t + b_2 t^2$$



Upward quadratic trend

Non-seasonal Decomposition Models with Trend - Method 2

Trend estimation by model fitting & removal

1. Generate synthetic data
2. Fit a linear regression
3. Fit a quadratic regression



Non-seasonal Decomposition Models with Trend

Combining Method 1 and 2

1. Generate synthetic data with quadratic trend
2. Differentiate
3. Observe the time-series still shows a trend
4. Detrend fitting a linear Regression



Decomposition Models with Trend and Seasonality

There are three basic methods for estimating/removing the trend and seasonal components:

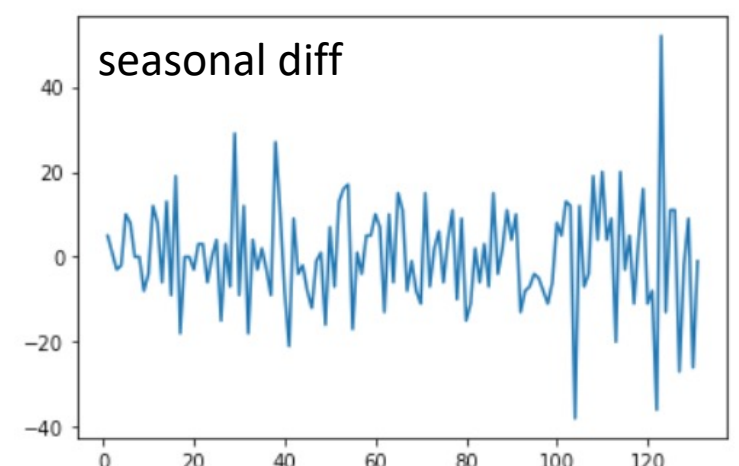
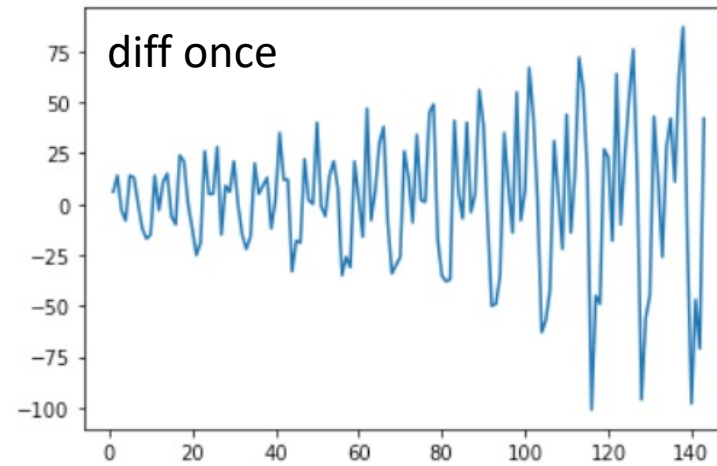
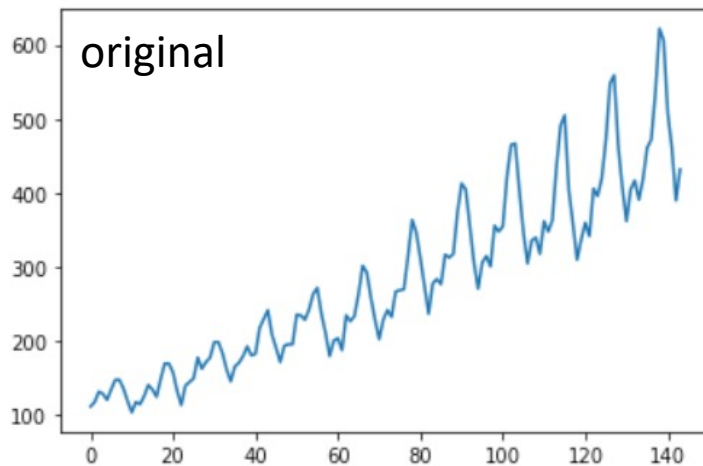
- Method 1 – Differencing
 - First we difference one or more time to remove the trend
 - Then we perform “**seasonal differencing**” to directly remove the season
- Method 2 - Filtering
 - First we estimate and remove the trend using a “**centered**” **moving average**
 - then we estimate and remove the seasonal component using “**periodic averages**”
- Method 3 - Joint-fit method
 - fitting a combined polynomial and **dynamic harmonic regression**

Decomposition Models with Trend and Seasonality - Method 1

Seasonal differencing

- Seasonal differencing of a time series $\{X_t\}$ in discrete time t given the seasonality's period d is the transformation of the series to a new time series $\{S_t\}$ where the values are the differences between the value of $\{X_t\}$ at time t and the the value of $\{X_t\}$ a period d before.

$$S_t = X_t - X_{t-d}$$



Decomposition Models with Trend and Seasonality - Method 1

Seasonal differencing

1. Generate synthetic data using a sine form
2. Apply "seasonal differencing"
3. Observe you removed the seasonality



Decomposition Models with Trend and Seasonality - Method 2

Estimate the trend using “centered” moving average

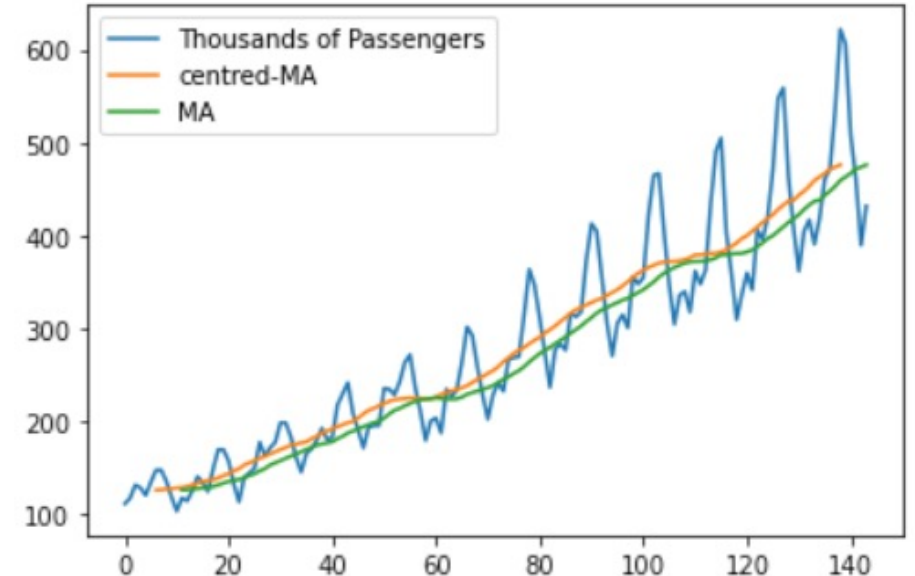
- Given the seasonality's period d
 - If d is even, the «centered» moving average is defined as

$$\widehat{m}_t = (0.5x_{t-q} + x_{t-q+1} + \dots + x_{t+q-1} + 0.5x_{t+q})/d$$

- If d is odd, the «centered» moving average is defined as

$$\widehat{m}_t = (x_{t-q} + x_{t-q+1} + \dots + x_{t+q-1} + x_{t+q})/d$$

- Notes:
 - there are no values for either the first q or the last q data points, because we do not have enough observations on either side to define the moving average for those values of t .
 - This «centered» moving average is different from the «normal» moving average



Decomposition Models with Trend and Seasonality - Method 2

Estimate the seasonal component

- To estimate the seasonal component using “**periodic averages**”
 1. Divide the detrended value in seasons of length d
 2. Compute the seasonal component values w_k by averaging each of the d points of the season (i.e., $k = 1, \dots, d$)
 3. Compute adjusted the seasonal component values s_k to ensure that they add to zero

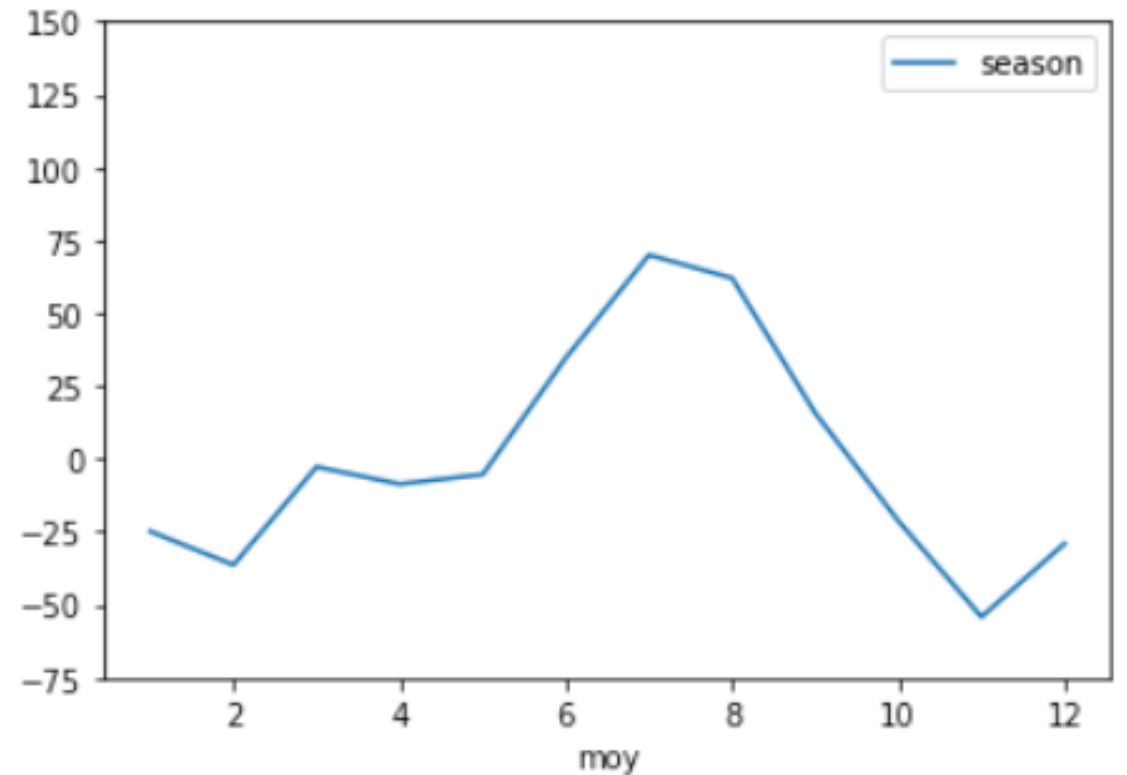
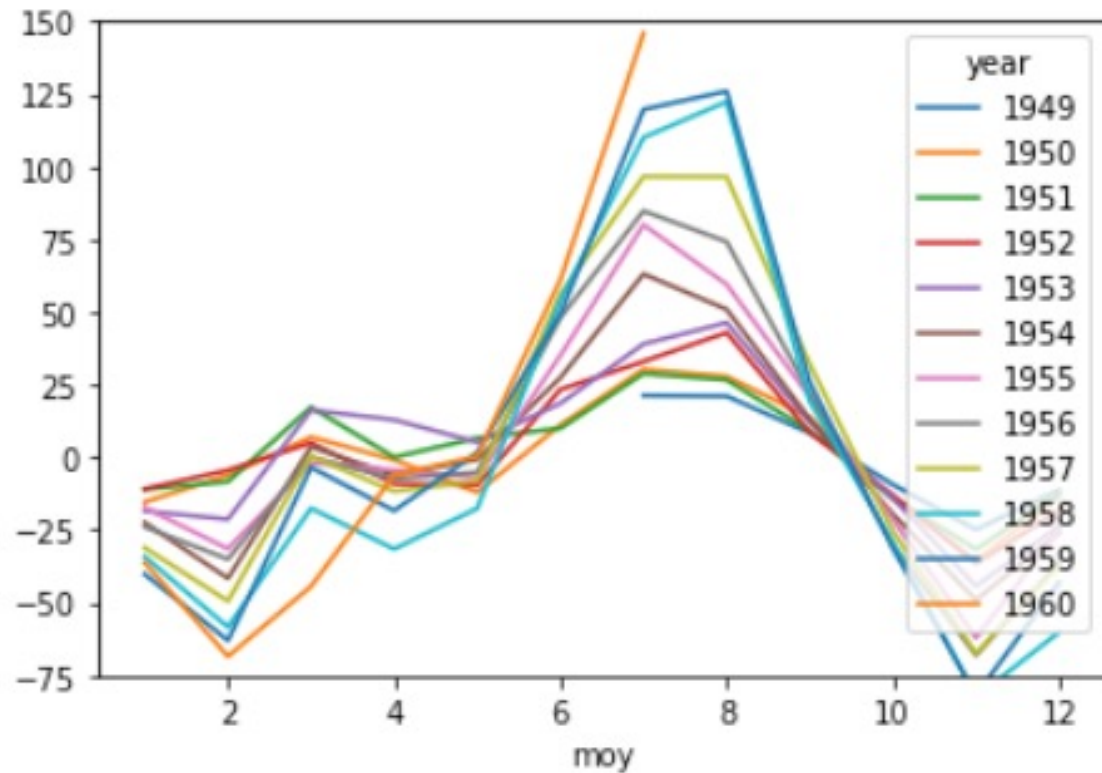
$$\hat{s}_k = w_k - d^{-1} \sum_{j=1}^d w_j, \quad k = 1, \dots, d$$

4. String together the adjusted seasonal component values in a sequence
5. Replace the sequence for each season

Decomposition Models with Trend and Seasonality - Method 2

Estimate the seasonal component

- The case of air travel time-series



Decomposition Models with Trend and Seasonality - Method 2 Filtering

1. Load the time series
2. estimate the trend by the “centered” moving average
3. Remove the trend from the time series
4. Estimate the seasonal component using “**periodic averages**”
5. Remove it from the detrended series



A detail I deliberately omitted

- In a multiplicative model, the components are multiplied:

$$X_t = m_t \times s_t \times Y_t$$

- Q: How do we differentiate a multiplicative time series?

A detail I deliberately omitted

- In a multiplicative model, the components are multiplied:

$$X_t = m_t \times s_t \times Y_t$$

- Q: How do we differentiate a multiplicative time series?
- A:

1st transform the multiplicative relationships into additive ones taking the logarithm:

$$\begin{aligned}\log(X_t) &= \log(m_t \times s_t \times Y_t) \\ \log(X_t) &= \log(m_t) + \log(s_t) + \log(Y_t)\end{aligned}$$

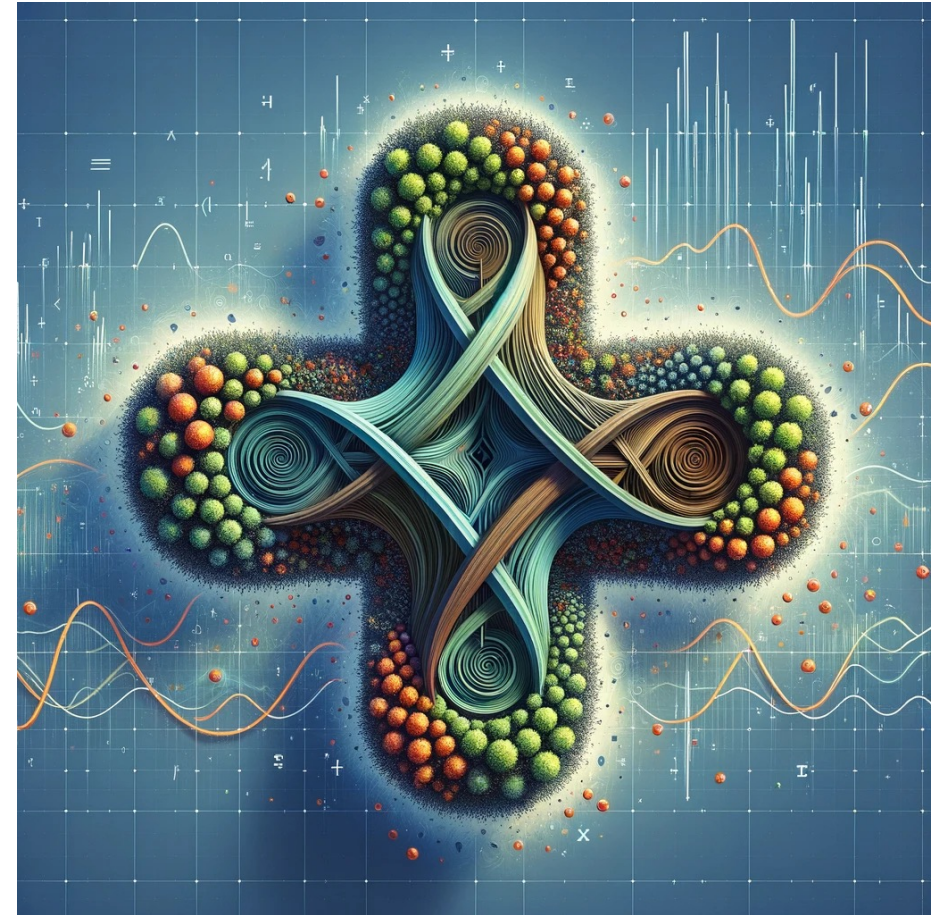
2nd you apply differencing to the logged series:

$$\begin{aligned}\log(m_t) &= \log(X_t) - \log(X_{t-1}) \\ \log(s_t) &= \log(X_t) - \log(X_{t-d})\end{aligned}$$

It's the math :-P

The log of the products
is the sum of the logs

Thus, the difference in the logs
is the log of the ratios



Decomposition Models with Trend and Seasonality - Method 3

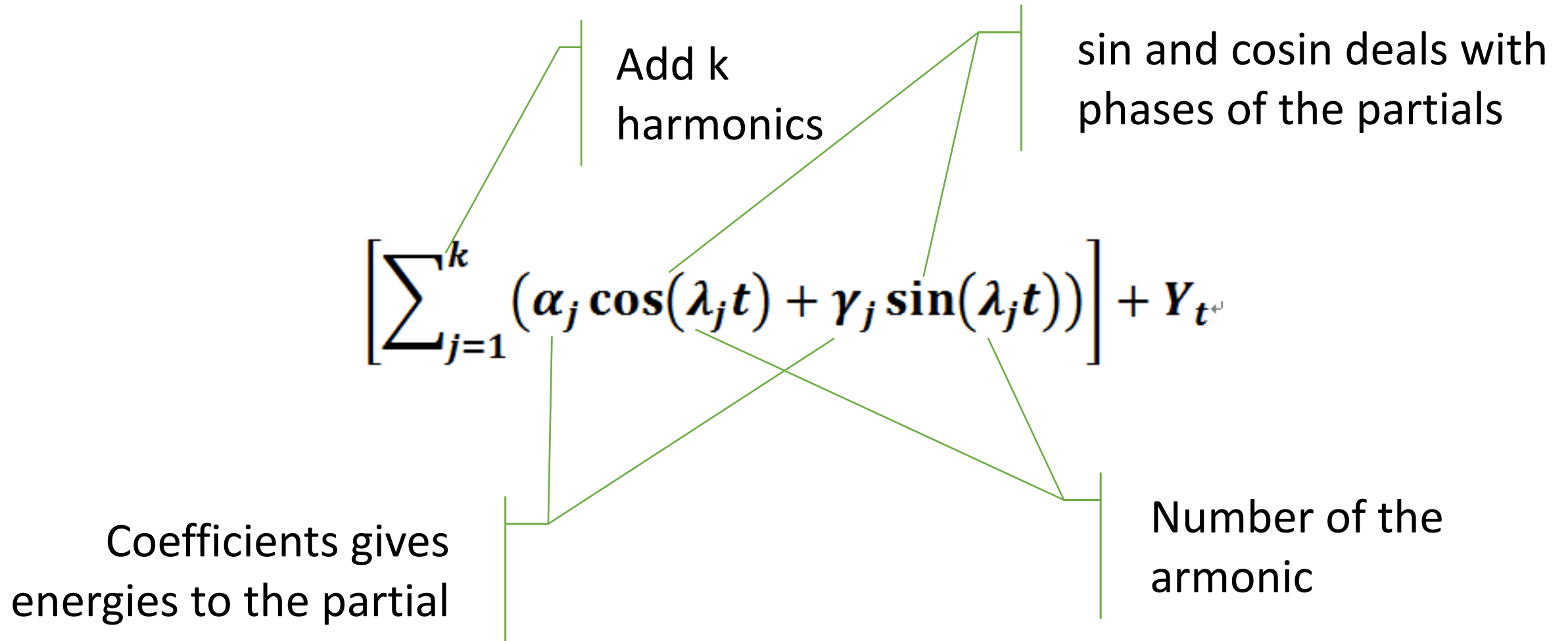
Joint-fit method

- we can fit a combined polynomial linear regression and harmonic functions to estimate and then remove the trend and seasonal component simultaneously as the following

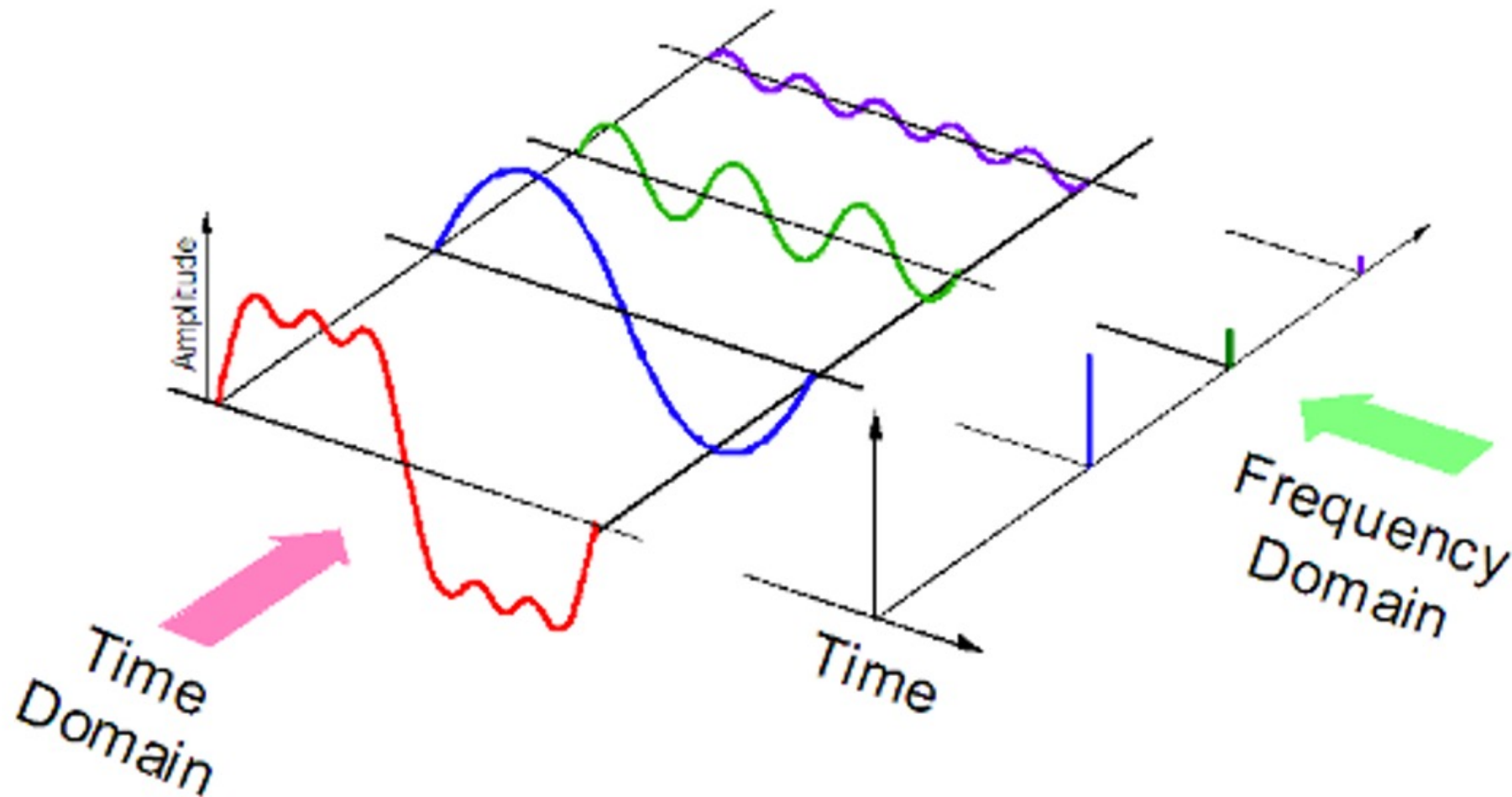
$$\begin{aligned} X_t &= m_t + s_t + Y_t \\ &= (\beta_0 + \beta_1 t + \beta_2 t^2) + \left[\sum_{j=1}^k (\alpha_j \cos(\lambda_j t) + \gamma_j \sin(\lambda_j t)) \right] + Y_t \end{aligned}$$

Decomposition Models with Trend and Seasonality - Method 3

Anatomy of the harmonic function



Decomposition Models with Trend and Seasonality - Method 3 Intuitively



Time-series Analytics Decomposition & Detrending

Giacomo Ziffer

giacomo.ziffer@polimi.it

PhD student @ Politecnico di Milano

Founder & CEO @ motus ml



POLITECNICO
MILANO 1863